



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY GUWAHATI

**Course Structure and Syllabus
(From Academic Session 2018-19 onwards)**

**B.TECH
CHEMICAL ENGINEERING
7TH SEMESTER**



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure

(From Academic Session 2018-19 onwards)

B.Tech 7th Semester: Chemical Engineering
Semester VII/ B. TECH/ChE

Sl. No	Sub-Code	Subject	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	CHE181701	Process Equipment Design-I	3	2	0	5	30	70
2	CHE181702	Polymer Science Engineering	3	0	0	3	30	70
3	CHE1817PE3*	Programme Elective – 3	3	0	0	3	30	70
4	HS181704	Principles of Management	3	0	0	3	30	70
Practical								
1	CHE181722	Project - I	0	0	8	4	50	100
2	SI181721	Internship-III (SAI- Industry)	0	0	0	2	-	200
TOTAL			12	2	8	20	170	580
Total Contact Hours per week : 22								
Total Credits: 20								

PROGRAMME ELECTIVE – 3 SUBJECTS		
Sl. No	Subject Code	Subject
1	CHE1817PE31	Biochemical Engineering
2	CHE1817PE32	Nanomaterials
3	CHE1817PE3*	Any other subject offered from time to time with the approval of the University

Course Code	Course Title	Hours per week L-T-P	Credit C
CHE181701	Process Equipment Design-I	3-2-0	5

MODULE 1: Design of Heat Transfer Equipment

Design and Drawing of Heat Transfer Equipment such as heat exchangers with and without phase change, evaporators, Condensers

MODULE 2: Design of High Pressure Systems

Design of high pressure vessels

MODULE 3: Design of Storage Tank

Design of storage tanks, head design and compensation design

MODULE 4: Drawing of Phase Separation Equipment

Drawing of physical separation equipment such as hydro-cyclones, packed towers, plate columns, electro static precipitators, schematic of pipe fitting and joints

MODULE 5: Design of Reactor Vessel :

Classification, Material of construction, Heating system and design considerations.

Textbooks/ Reference Books:

1. D.Q. Kern, Process Heat Transfer, 2nd Ed., Tata McGraw - Hill, 1997.
2. L. E. Brownell and E.H. Young, "Process Equipment Design - Ves Design", Wiley Eastern Edn. New York, 1968. 2. R. H. Perry, "Chemical Engineers' Handbook", 7th Edn., McGraw Hill, N York, 1998.
3. M. V. Joshi, "Process Equipment Design", 2nd Edn. Mac Millan Press, N Delhi, 1996.
4. Process Equipment Design-mechanical aspects; B C Bhattachrjee
5. J. M. Coulson and J. F. Richardson, "Chemical Engineering.", Vol-VI, Pergam Press, New York, 1987.

Course Code	Course Title	Hours per week L-T-P	Credit C
CHE181702	Polymer Science Engineering	3-0-0	3

MODULE 1: Basics

Formation, Classification, Methods of Polymer formation, Step Growth, chain growth, Z-N catalyst

MODULE 2: Molecular Weight and Molecular Weight Distribution

MODULE 3: Structure of Polymer Molecules

Based on chemical composition – homo and co-polymer, Structural Model-Random Coil, fringed micellar model, spherulites

MODULE 4: Physical States and Transitions

Amorphous and crystalline, Transition temperature-glass transition temperature, melting temperature

MODULE 5: General Properties and Testing of Polymers

MODULE 6: Polymer Processing/Fabrication

Polymer additives, Injection molding (plunger and screw type) Compression molding, Extrusion – flat, sheet and tubing Pultrusion, Blow molding, Foams, thermoforming, vacuum forming Spinning, melt, wet and dry

MODULE 7: Finishing Processes

MODULE 8: Application, Recycling, Reuse and Reclaim

MODULE 9: Rheology

Viscous flow, models of Newtonian and Non-Newtonian, Visco-elasticity – Maxwell Model

MODULE 10: Introduction to Rubber Elasticity

MODULE 11: Introduction to Nano-Composites

MODULE 12: Inhouse Demonstration of Blow Moulding & Injection Moulding Machine

Textbooks/ Reference Books:

1. Polymer science by V.R. Gowarikar.
2. Sinha, Outlines of Polymer Technology, Manufacture of Polymers, Prentice Hall India
3. Fundamentals of Polymers: Raw Materials to finished Products by Narayan Karak, PHL
4. Fried, Polymer Science and Technology, 2nd Ed, Prentice Hall I

Course Code	Course Title	Hours per week L-T-P	Credit C
CHE1817PE31	Biochemical Engineering	3-0-0	3

MODULE 1: Application of Bioprocess in Various Fields

MODULE 2: Chemicals of Life

Introduction and basic functions of - Lipids; Sugar & Polysaccharides; - Nucleotides & RNA & DNA; Amino acid & Proteins; Hybrid bio-chemicals.

MODULE 3: Simple Enzyme Kinetics with One and Two Substrates

Michaelis – Menten kinetics, Kinetics for reversible reactions. Substrate activation & inhibition. Multiple substrate on single enzyme. Influence on enzyme activity. Enzyme deactivation.

MODULE 4: Application of Enzyme as Catalysts

Hydrolysis of starch and cellulose by hydrolytic enzyme.; Medical application; Non hydrolytic enzyme.

MODULE 5: Immobilized Enzyme Technology -Brief overview.

MODULE 6: Basic Fermentation Engineering

Media preparation, culture, sterilization; - Manufacture of biochemical products

MODULE 7: Transport Phenomena in Bio Process

Gas liquid transfer in cellular system; Heat transfer.

MODULE 8: Kinetics in Cell Culture

Reaction Kinetics of balance growth, Death kinetics.

MODULE 9: Bioreactors

Introduction, Basic features and types

MODULE 10: Downstream Processing

Bio-product recovery and bio-separations

Textbooks/ Reference Books:

1. Biochemical Engineering by Bailey, McGraw Hills.
2. Bioprocess Engineering-Basic Concepts by M L Shuler and F Kargi, Prentice Hall India
3. Biochemical Engineering by Mukesh Doble, S. Gummadi, Prentice Hall India
4. Biochemical Engineering: Principles and Concepts, S. T. A. Inamdar, Prentice Hall India.

Course Code	Course Title	Hours per week L-T-P	Credit C
CHE1817PE32	Nanomaterials	3-0-0	3

Course Outcomes:

After completing the course, the students will be able to

CO1: Have a systematic overview of Nanomaterials in terms of classifications, properties, storage and precautions in handling.

CO2: Adopt different methods to fabricate Nanomaterials from different synthetic and biobased sources.

CO3: Apply different tools to characterize Nanomaterials.

CO4: Apply the acquired knowledge to develop Nanomaterials for specific applications.

CO5: Understand the future prospects of Nanomaterials in current environmental scenario.

MODULE 1: Fundamentals of Nanomaterials

Historical development of nanomaterials, Classification, Size & Scale, Structure and Bonding in Nanomaterials, Basics of Fabrication Methods, Preparation, Safety, Handling and Storage issues.

MODULE 2: Types of Nanostructure and Properties of Nanomaterials

One dimensional, Two dimensional and Three dimensional nanostructured materials, Thermal, Magnetic, Optical, Electrical and mechanical properties of nanomaterials, Quantum Dots, Carbon dots, Carbon nano tubes, metal oxides, Nanomaterials from biobased sources, nanocomposites.

MODULE 3: Nanotechnology Utilized in Different Materials

Introduction, Nanotechnology in polymers, ceramics, fibres and metals, techniques of fabrication.

MODULE 4: Characterization tools for Nanomaterials

X-Ray Diffraction (XRD), Scanning Electron Microscopy, Transmission Electron Microscopy, Atomic Force Microscopy, UV/Visible Spectroscopy, Delsa nano.

MODULE 5: Applications of Nanomaterials

Nano Sensor, Nano catalysts, Water Treatment and the Environment, Paints, Food and Agriculture Industry, Cosmetics and Consumer Goods.

Textbooks/ Reference Books:

1. Nanostructures and Nanomaterials: Synthesis, Properties and Applications by G. Cao, Imperial College Press, 2004.
2. Nanoscale Science and technology by Robert Kelsall (editor), Ian W. Hamley (co-editor), Mark Geoghegan (co-editor), ISBN: 978-0-470-85086-2
3. The Chemistry of Nanomaterials: Synthesis, Properties and Applications by C. N. R. Rao, A.Muller, A. K. Cheetham, WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim, ISBN: 3-527-30686-2.
4. Nanoscale Materials in Chemistry Edited by Kenneth J. Klabunde, John Wiley & Sons, Inc., ISBNs: 0-471-38395-3 (Hardback); 0-471-22062-0.
5. Textbook of Nanoscience and Nanotechnology, B.S. Muty, P. Shankar, Baldev Raj, B.B Rath and James Murday, University Press, IIM (ISBN-978 81 7371 738 3).
6. Introduction to Nanotechnology by Charles P. Poole Jr and. Frank J. Owens, Wiley-Inter science, 2003

Course Code	Course Title	Hours per week L-T-P	Credit C
HS181704	Principles of Management	3-0-0	3

MODULE 1: Introduction

(6 Lecture)

Definition and meaning of management, Characteristics of management, importance of management, functions of management-planning, organising, directing, staffing, coordination and controlling etc., principles of management, Difference between administration and management

MODULE 2: Financial Management

(6 Lecture)

Definition and management of financial planning, importance and characteristics of sound financial plan, concepts of capital- fixed capital and working capital, source of finance, fund flow statement.

MODULE 3: Marginal Costing

(6 Lecture)

Definition and meaning of marginal costing, advantages, marginal cost equation, contribution, profit-volume ratio, break even analysis, margin of safety.

MODULE 4: Cost Accounting

(6 Lecture)

Cost Accounting- Concept and benefit, elements of cost, preparation of cost sheet with adjustment of raw materials, work-in-progress and finished goods.

MODULE 5: Capitalisation

(4 Lecture)

Definition and meaning of capitalisation, over and under capitalisation.

MODULE 6: Motivation

(6 Lecture)

Introductory observation, definition of motivation, motivational technique, features of sound motivational system.

MODULE 7: Leadership

(6 Lecture)

Concept of leadership, principles of leadership, functions of leadership, qualities of leadership, different styles of leadership.

Textbooks/Reference Books:

1. Principle of Business Management: RK Sharma, Shashi K.Gupta
2. Business Organisation and Management: SS Sarkar, RK Sharma, Shashi K.Gupta
3. Industrial Organisation and Management: SK Basu, KC Sahu, B Rajviv
4. Principles of Management by Dr. A. K. Bora: Chandra Prakash, Guwahati.
5. Management Accounting: RK Sharma, Shashi K Gupta
6. Cost Accounting: SP Jain, K I Narang
7. Cost Accounting, RSN Pillai, V Bhagawati
8. Principles of Management: RN Gupta
9. Principles of Management: RSN Pillai, S. Kala
10. Principles of Management: Dipak Kumar Bhattacharjee

Course Code	Course Title	Hours per week L-T-P	Credit C
CHE181722	Project – I	0-0-8	4

Course Code	Course Title	Hours per week L-T-P	Credit C
SI181721	Internship-III (SAI-Industry)	0-0-0	2
